Lecture 23

Thursday, 27 April 2023 10:30 AM

→ HW's: Can ask for solutions

→ Course eval → Friday deadline

$0.1\ eV \sim 10 k_B T$ (at 1 atm)

Zeolite continuation: DEE formation (Diethyle ether)

Microkinetic model → degree of rate control (sensitivity analysis)

(Using higher accuracy functional to get better elementary step endothermic value)

} K's changed a lot? → Maybe other steps were RDS / imp.? → Need to compare to the experimental values/data

MARI would depend on changing T, P & reactor conditions

Review of surface and gas phase chemical potential analysis

① (Ideal) gas treatments: divide them into translational, rotational, vib, electronic (DFT) parts

$$q_{vib} = \prod_i^{\text{mode}} \frac{e^{-h\nu_i\beta/2}}{1 - e^{-h\nu_i\beta}}$$ (per molecule) — includes ZPE

$$q_{rot} = \frac{I}{\sigma \beta \hbar^2}$$ (linear) per molecule

$$q_{trans} = \left(\frac{2\pi m k_B T}{h^2}\right)^{3/2} V$$ (per molecule)

$$q_{el} = e^{-E_{DFT}/k_B T}$$

$$Q_{tot} = \frac{q_{trans}^N}{N!} q_{vib}^N q_{rot}^N e^{-E_{DFT}/k_B T}$$ (E_{DFT}: total energy)

$$F_{tot} = -k_B T \ln Q_{tot}, \quad \mu = \frac{\partial F_{tot}}{\partial N} = -k_B T \frac{\partial (\ln Q_{tot})}{\partial N}$$

$\ln N! \approx N \ln N - N$

$$\mu = -k_B T \ln\left[\left(\frac{2\pi m k_B T}{h^2}\right)^{3/2}\left(\frac{V}{N}\right)\right] - k_B T \ln q_{vib} - k_B T \ln q_{rot} + \frac{\partial E_{DFT}}{\partial N}$$

$\left(\frac{V}{N}\right) = P/k_B T$

Standard state: $\mu^\circ = \mu(P = P^\circ)$

$$\mu^\circ = -\frac{3}{2} k_B T \ln\left(\frac{2\pi m k_B T}{h^2}\right) - k_B T \ln\left(\frac{P^\circ}{k_B T}\right) - k_B T \ln q_{vib} - k_B T \ln q_{rot} + \frac{\partial E_{DFT}}{\partial N}$$ (includes ZPE) } Use to calculate K's in microkinetic model

$$\therefore \mu = \mu^\circ + k_B T \ln\left(\frac{P}{P^\circ}\right)$$

$$\mu^\circ = \varepsilon_{DFT} + ZPE - T\bar{S}, \quad \bar{S} = \frac{\partial S}{\partial N}$$

$$\frac{\partial E_{DFT}}{\partial N} = \frac{\partial}{\partial N}(N \varepsilon_{DFT}) = \varepsilon_{DFT}$$ (per molecule)

$$\frac{\partial E_{DFT}}{\partial N} = \frac{\partial}{\partial N}\left(N\,\varepsilon_{DFT}\right) = \varepsilon_{DFT} \quad \text{(per molecule)}$$

(2) Liquid phase treatments – complicated

(3) Surface processes:

(a) First order treatment: calculate harmonic modes on surface ($3N_{atoms}$ modes) and use harmonic partition function:

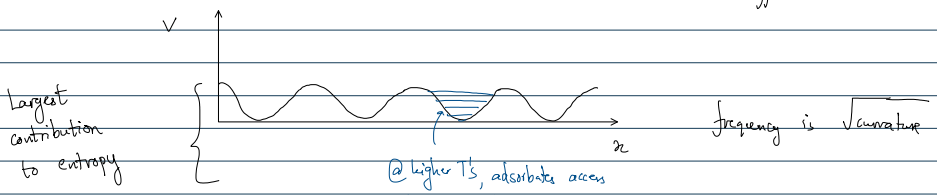
$$Q_{tot} = q_{vib}^{N}\, e^{-E_{DFT}/k_BT}\, \frac{N_s!}{N!\,(N_s-N)!}$$

$$\mu = \frac{\partial E_{tot}}{\partial N} = -k_BT \ln\left[\prod_{i}^{modes} \frac{e^{-h\nu_i\beta/2}}{1-e^{-h\nu_i\beta}}\right] + \frac{\partial E_{DFT}}{\partial N} + k_BT \ln\left(\frac{\theta}{\theta_*}\right); \quad \theta_* = 1-\theta_A-\theta_B-\theta_C$$

μ° (includes ZPE); activity

$$\therefore \boxed{\mu = \mu^\circ + k_BT \ln\left(\frac{\theta}{\theta_n}\right)}$$; For TS, do not include reaction coordinate

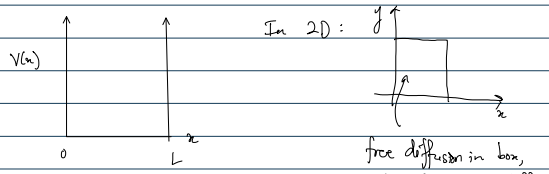
Caution: This approach fails for low frequency modes ($\nu_i \lesssim 150\,cm^{-1}$). } Low frequencies imply low diffusion barriers – anharmonic



Crude correction: take all modes w/ $\nu_i \lesssim 150\,cm^{-1}$, and assume entropies are all equal to a mode w/ $\nu = 150\,cm^{-1}$ (!?)

(b) Better treatments of low freq. modes

(i) Particle-in-a-box (for adsorbate)



In 2D:

free diffusion in box, but molecules are still localized.

In 2D, $E_{n,l} = \frac{\pi^2\hbar^2}{2m}\left[\frac{n^2}{L_x^2} + \frac{l^2}{n_y^2}\right]$ $n,l = 1,2,\ldots$

$$q_{trans}\ (\text{per molecule}) = \sum_{n=1}^{\infty}\sum_{l=1}^{\infty} \exp\left[\frac{-\pi^2\hbar^2}{2mk_BT}\left(\frac{n^2}{L_x^2}+\frac{l^2}{L_y^2}\right)\right]$$

$$\approx \int_0^{\infty}\int_0^{\infty} e^{\frac{-\pi^2\hbar^2 n^2}{2mk_BTL^2}} e^{\frac{-\pi^2\hbar^2 l^2}{2mk_BTL^2}}\, dn\, dl \quad ; \text{ if } L_x = L_y = L$$

$$= \frac{2\pi mk_BT}{h^2} A_{box} \qquad A_{box} = L^2 = \text{area of free diffusion for each adsorbate}$$

$$Q_{trans} = q_{trans}^{N} \quad ; \quad \mu_{trans} = -k_BT \ln\left[\frac{2\pi mk_BT}{h^2} A_{box}\right]$$

define this for the area that you care around the active site

$$\boxed{\mu^\circ = -k_BT \ln\left[\frac{2\pi mk_BT}{h^2} A_{box}\right]}$$

Replace low frequency trans modes w/ this
everything else is same

Useful for: zeolites, surfaces @ high coverages

(ii) Two-dimensional ideal gases (simple adsorption/desorption, weakly adsorbates)

2D ideal gases: $q_{trans} = A\frac{2\pi mk_BT}{h^2}$ (per molecule)

$$Q_{tot} = \frac{q_{trans}^{N}}{N!} q_{vib}^{N} q_{rot}^{N} e^{-E_{DFT}/k_BT}$$

↖ Identify for low freq. modes & high freq. modes

gives $\frac{A}{N} = \frac{a_s}{\theta}$; $\theta = \frac{N}{N_s}$

$$\mu = -k_BT \ln\left[\frac{2\pi mk_BT}{h^2} a_s\right] - k_BT \ln(q_{vib}\, q_{rot}) + \frac{\partial E_{DFT}}{\partial N}$$

↳ includes ZPE

$+k_BT \ln\theta$

$$\boxed{\mu = \mu^\circ + k_BT \ln\theta}$$

↖ No θ_* because i.g. have no volume (so no site blocking term)

→ Not correct for real molecules @ high coverages

→ still approx. true if $\boxed{\theta_* \approx 1}$; $\mu = \mu^\circ + k_BT \ln\left(\frac{\theta}{\theta_*}\right)$

Rough correction:

change $A \to A - N a_{ads}$

($N a_{ads}$: excluded from the area)

then $\frac{A}{N} \to \frac{A - N a_{ads}}{N} = \frac{N_s a_s - N a_{ads}}{N}$

$$= a_s \frac{\left(1 - \frac{a_{ads}}{a_s}\theta\right)}{\theta}$$

$\Rightarrow \mu^\circ$ is the same,

$$\mu = \mu^\circ + k_B T \ln \frac{\theta}{\theta_*} \; ; \; \theta_* = 1 - \frac{a_{ads}}{a_s}\theta$$

If the coverage is getting too high ($\theta_* < 1$), consider particle in a box (or diff. model))